

## OBJECTIVE

Gold-medalist Software Engineering graduate with hands-on research in computer vision, vision-language models, and medical imaging. I aim to make large vision models faster, lighter, and more trustworthy — through token compression, knowledge distillation, and interpretable deep learning — so they can run reliably in real-world and resource-limited settings.

## EDUCATION

Oct 2022 – **B.S. Software Engineering — Gold Medalist, CGPA 3.92/4.00**  
Jun 2026 *University of Haripur, KP, Pakistan*  
*Thesis: Augmented Reality & Deep Learning Assisted Enhanced Museum Experience*

## RESEARCH INTERESTS

Efficient Deep Learning · Interpretable AI for Medical Imaging · Computer Vision · Self-Supervised Representation Learning · Image Recognition & Object Detection · Vision-Language Models

## MAJOR RESEARCH PROJECT

2025 – 2026 **Augmented Reality & Deep Learning Assisted Enhanced Museum Experience**  
*Final Year Research Project (Thesis) — University of Haripur*

- Built a multimodal AI system integrating computer vision, deep learning, speech interaction, and multilingual assistance for real-world cultural heritage recognition (case study: Taxila Museum).
- Curated a custom dataset of Taxila Museum artifacts; performed end-to-end data collection, preprocessing, and augmentation pipelines; developed vision-language recognition pipelines for low-resource artifact categories.
- Research paper on hybrid vision-language recognition for cultural heritage systems under preparation for publication.
- Nominated by the Directorate General of Science & Technology (DoST), KP for research funding, recognizing scientific merit and societal impact.
- Awarded Top Project at the Final Year Project Exhibition, University of Haripur.

## ACADEMIC PROJECTS

### Explainable AI for Knee Osteoarthritis Detection & Severity Grading — Python, TensorFlow, DenseNet-121

- DenseNet-121 ensemble that grades knee osteoarthritis into 5 severity levels from X-rays, reaching 98.75% accuracy, with Grad-CAM heatmaps showing which regions drive each prediction.

### PulmoVision: Automated Pneumonia Diagnosis Using Deep Learning — Python, TensorFlow, ResNet

- ResNet transfer learning pipeline for pneumonia classification from chest X-rays, deployed as a public web app on Streamlit Community Cloud.

[Live App](#) · [Code](#)

### Performance Benchmarking of Machine Learning Algorithms — Python, scikit-learn, Pandas

- Benchmarking multiple machine learning algorithms across datasets and problem instances, analyzing performance variations to identify relationships between dataset characteristics and algorithm effectiveness.

### NailScan: AI-Powered Nail Condition Analyzer — Python, PyTorch, YOLOv5, OpenCV

- Built and deployed a nail health screening app: a YOLOv5 segmentation model locates the nail, then a ResNet-152 classifier grades it into one of 6 conditions.

[Live App](#) · [Code](#)

### Encoder-Aware Visual Token Compression for Efficient VLMs — Python, PyTorch, ViT

- Implementing adaptive, encoder-dependent visual token compression for vision-language models, benchmarking a random baseline and Top-K pruning against an adaptive compression method on ViT.

[Code](#)

### Data Structures & Algorithms Visualization Suite — C++

- Desktop application with graphical visualization of sorting, tree traversals, and graph pathfinding algorithms.

## TECHNICAL SKILLS

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|-------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Languages</b>  | Python, JavaScript, C++                                                                                                  |
| <b>Frameworks</b> | PyTorch, TensorFlow, Keras, Hugging Face Transformers, OpenCV, Scikit-learn, FastAPI, Pandas, Seaborn, Plotly, Streamlit |
| <b>Dev Tools</b>  | Git/GitHub, MySQL, SQL, Linux/Bash, REST APIs, OOP, Web Development                                                      |

## EXPERIENCE

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| Jul 2026 – Present  | <b>AI Intern</b><br><i>UAVs Lab, National Radio and Telecommunication Corporation (NRTC) — Haripur, Pakistan</i> <ul style="list-style-type: none"><li>Working on an object detection and pattern recognition module for a UAV-based mission: comparing before/after aerial imagery from a drone's outbound and return flight path to identify and flag missing objects.</li><li>Collaborating within a larger multi-module team project, owning the detection and comparison component end-to-end.</li></ul> |
| Jul 2025 – Aug 2025 | <b>IT Directorate Intern</b><br><i>Senate of Pakistan — Islamabad, Pakistan</i> <ul style="list-style-type: none"><li>Contributed to software development and web-based systems within the IT Directorate.</li><li>Assisted in maintaining and troubleshooting internal web applications used across departments.</li><li>Collaborated with senior IT staff on requirement gathering and testing for ongoing system upgrades.</li></ul>                                                                       |
| Jul 2024 – Aug 2024 | <b>Game Development Intern</b><br><i>Codematics Inc. — Abbottabad, Pakistan</i> <ul style="list-style-type: none"><li>Developed 2D/3D Unity projects, including animations and shader graphs.</li><li>Explored AI-driven gameplay elements and prototyped interactive mechanics.</li><li>Collaborated with a small team on game design iteration and playtesting feedback.</li></ul>                                                                                                                          |

## AWARDS & ACHIEVEMENTS

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**Gold Medal**, BS Software Engineering (CGPA 3.92/4.00), University of Haripur — Jun 2026  
**Top Project**, Final Year Project Exhibition, University of Haripur — Jul 2026  
**Laptop Recipient**, PM Youth Program (Academic Excellence) — Nov 2023  
**Top Performer**, Python Frenzy Competition, IT Club — Sep 2024  
A1 grades in SSC & FSc (2019–2021); Top 10, Abbottabad Board (FSc); 93rd percentile, Engineering/IT entry test

## CONFERENCES & SEMINARS

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AI in Healthcare Webinar — Abu Dhabi, UAE  
1st International Conference on Computational Sciences and Innovations (ICCSI) — Haripur, Pakistan  
CyberFest — Peshawar, Pakistan

## CERTIFICATIONS

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- [AI in Healthcare Specialization](#) — Stanford University
- [Deep Learning for Computer Vision: Techniques and Applications](#) — Khalifa University
- [Computer Vision Specialization](#) — University of Colorado Boulder
- Modern AI Models for Vision and Multimodal Understanding — University of Colorado Boulder
- [Multimodal Intelligence: Vision, Audio and Language in Action](#) — Coursera
- [Unsupervised Learning, Recommenders, Reinforcement Learning](#) — DeepLearning.AI
- [Fundamentals of Building AI Agents](#) — IBM
- [Explainable AI: Scene Classification and Grad-CAM Visualization](#) — Coursera
- [Interpretable Machine Learning](#) — Duke University
- [Build Multimodal Generative AI Applications](#) — IBM
- AI in Healthcare Webinar — UAEU
- Crash Course on Python — Google

## LEADERSHIP & SERVICE

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| Jun 2025 – Present | Vice President, IT Computing Society, University of Haripur |
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| Dec 2025 | Career Expo 2025 Organizer, University of Haripur <a href="#">[Certificate]</a>     |
| Apr 2025 | Event Manager, 4th Convocation, University of Haripur <a href="#">[Certificate]</a> |
| Jan 2024 | Team Lead, Social Work Visit — Aghosh Al-Khidmat, Haripur <a href="#">[Photos]</a>  |

## LANGUAGE SKILLS

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Urdu — Mother tongue | English — Fluent

## REFERENCES

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**Dr. Dawar Khan**

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